THIN LENSES AND RAY TRACING

1. Thin Lens

A lens is a transparent optical medium bounded by two surfaces, at least one of which is curved. A thin lens is a lens whose thickness is small compared with its radii of curvature.

A lens forms an image by refracting light rays passing through it.

2. Types of Lenses

There are two main types of lenses:

1. Convex Lens (Converging Lens)

A convex lens is thicker at the centre and thinner at the edges. It converges parallel light rays to a focal point.

- It has a positive focal length.
- It can form real or virtual images.
- It is used in magnifying glasses, cameras and microscopes.

2. Concave Lens (Diverging Lens)

A concave lens is thinner at the centre and thicker at the edges. It diverges parallel light rays so that they appear to come from a focal point.

- It has a negative focal length.
- It forms a virtual, erect and diminished image for a real object.
- It is used in spectacles for correcting myopia.

3. Important Terms

1. Principal Axis: The straight line passing through the optical centre and the centres of curvature of the lens surfaces.
2. Optical Centre (O): The point in a thin lens through which a ray passes approximately without deviation.
3. Principal Focus (F): The point where rays parallel to the principal axis converge, or appear to diverge from, after refraction.
4. Focal Length (f): The distance between the optical centre and the principal focus.
5. Object Distance (u): The distance of the object from the optical centre.
6. Image Distance (v): The distance of the image from the optical centre.
7. Magnification (m): The ratio of image height to object height.

4. Thin Lens Formula

The thin lens formula gives the relationship between focal length, object distance and image distance.

Formula:

$1/f = 1/v - 1/u$

Where:

- f = Focal length
- u = Object distance
- v = Image distance

This formula is used with the Cartesian sign convention.

5. Magnification of a Lens

Magnification indicates how large or small an image is compared with the object.

Formula:

$m = h_i/h_o = v/u$

Where:

- m = Magnification
- h_i = Image height
- h_o = Object height
- v = Image distance
- u = Object distance
- Positive magnification indicates an erect image.
- Negative magnification indicates an inverted image.
- |m| greater than 1 indicates an enlarged image.
- |m| less than 1 indicates a diminished image.

6. Ray Tracing

Ray tracing is a method of determining the position, size and nature of an image by drawing the paths of selected light rays passing through a lens.

Rules of Ray Tracing

For a convex lens:

1. A ray parallel to the principal axis passes through the principal focus on the opposite side of the lens.
2. A ray passing through the optical centre travels approximately straight without deviation.
3. A ray passing through the principal focus on the object side emerges parallel to the principal axis.

For a concave lens:

1. A ray parallel to the principal axis diverges after refraction and appears to originate from the principal focus on the same side.
2. A ray passing through the optical centre travels approximately straight without deviation.
3. A ray directed towards the principal focus on the opposite side emerges parallel to the principal axis.

The image is located where the refracted rays meet or where their backward extensions meet.

7. Image Formation by a Convex Lens

Object Position	**Image Position**	**Nature of Image**
At infinity	At F_2	Real, inverted, highly diminished
Beyond $2F_1$	Between F_2 and $2F_2$	Real, inverted, diminished
At $2F_1$	At $2F_2$	Real, inverted, same size
Between F_1 and $2F_1$	Beyond $2F_2$	Real, inverted, enlarged
At F_1	At infinity	Image at infinity
Between F_1 and O	On the same side as the object	Virtual, erect, enlarged

8. Image Formation by a Concave Lens

Object Position	**Image Position**	**Nature of Image**
At infinity	At F_1	Virtual, erect, highly diminished
At any finite distance	Between O and F_1 on the object side	Virtual, erect, diminished

9. Power of a Lens

The power of a lens measures its ability to converge or diverge light.

Formula:

$P = 1/f$

Here, focal length must be expressed in metres.

- Unit of power: Dioptre (D)
- Convex lens: Positive power
- Concave lens: Negative power

For example, if the focal length is +0.5 m:

$P = 1/0.5 = +2$ D

10. Applications of Thin Lenses

1. Cameras and projectors.
2. Microscopes and telescopes.
3. Spectacles and contact lenses.
4. Magnifying glasses.
5. Optical instruments used in laboratories.

Conclusion

Thin lenses form images by refraction of light. Convex lenses converge light, whereas concave lenses diverge light. Ray tracing helps determine image position and nature, while the thin lens formula is used to calculate image distance and focal length.